

Lionheart of England in 1199. Initially, these early European models were drawn by the bowman bending forward, placing his foot in the stirrup at the end of the stock, and attaching the string to a hook on his belt; when he straightened his back, the string was drawn upwards, “spanning” the bow. From the 14th century onwards, however, mechanisms were introduced – first a windlass and pulley, and later a ratchet known as a cranequin. These systems allowed the use of composite and steel bows of even greater strength: the effect of these weapons on the battlefield was awesome.

ARCHERS AND PIKEMEN

The main drawback of the crossbow, however, was its slow rate of fire. One or two shots a minute was the best an experienced bowman could manage, and the crossbow was often a more effective weapon when used in sieges than on an open battlefield. The longbow, on the other hand, made a huge impact on European battlefields when deployed *en masse* by English armies in the 14th and 15th centuries. Preferably made of yew, the “self” bow (made from a single piece of wood) required constant practice, as well as great physical strength,

on the part of the archer. Its effectiveness depended upon an experienced archer who could loose around 17 arrows a minute. English kings discovered that by packing thousands of longbows onto the battlefield, they could produce a great density of fire – from a modern perspective, almost a similar effect to that of the machine-gun. The longbow saw spectacular successes in battle against the French, at Crécy in 1346, Poitiers in 1356, and Agincourt in 1415.

A demonstration of the importance of combining tactics with existing military technology was provided by the Swiss pikemen in the late 15th century. The long pike was a supremely simple weapon, but the Swiss developed a new method of using it offensively, grouping their infantry together and fighting as massed columns of pikemen advancing rapidly upon the enemy. Even more so than the introduction of gunpowder weapons, their victories in the 1470s set the scene for a new era in infantry warfare.



▲ SALLET

The sallet, a form of helmet introduced in the 15th century and widely used by infantry, gave excellent protection to the neck.



◀ BOWMEN IN PITCHED BATTLE

A representation of the Battle of Aljubarrota, in 1385, shows bowmen shooting at unrealistically close range. The armour pictured is more typical of the late 15th century.

KEY BATTLE

THE BATTLE OF BANNOCKBURN 24 JUNE 1314

Fought in June 1314, Bannockburn was a famous victory for Scottish foot soldiers, led by Robert Bruce, over King Edward II's mounted English knights. The Scots took up positions on a slope above a stream. Their unarmoured soldiers formed a schiltrion – a tight unit bristling with spears. The schiltrion held off repeated charges by the English knights, before eventually driving them from the field in disarray.



▲ A modern illustration gives an impression of the chaos of fallen horses and knights at Bannockburn.